Examiner only

10. Some students carried out an experiment to investigate the relative hardness of four water samples, **A**, **B**, **C** and **D**.

The students measured 20 cm^3 of sample **A** into a measuring cylinder. They added soap solution, 1 cm^3 at a time. After each addition the mixture was shaken. The volume of soap solution needed to produce 1 cm of lather was recorded.

Samples **B**, **C** and **D** were tested in the same way.

They then repeated each experiment with samples that had been boiled and others that had been treated with washing soda.

Their results are shown in the table.

Water sample	Volume of soap solution needed (cm^3)		
	Before boiling	After boiling	After washing soda
A	8	2	2
B	12	12	2
C	2	2	2
D	9	6	2

(a) Use the results in the table to answer parts (i) and (ii).

(i) Identify the type of hardness found in sample **A**.

Give a reason for your answer. [2]

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(ii) Identify the sample containing both temporary and permanent hardness.

Give a reason for your answer. [2]

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(b) Another group of students carried out the same experiment on the same four water samples.

However, this group measured smaller volumes of each sample into the measuring cylinder. Their results are shown in the table.

Water sample	Volume of soap solution needed (cm^3)		
	Before boiling	After boiling	After washing soda
A	5	1	1
B	8	6	1
C	1	1	1
D	7	4	1

Compare the conclusions that can be drawn by the two groups. [2]

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(c) Permanent hardness is removed by ion exchange.

Explain how ion exchange works. [2]

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(d) Magnesium ions cause hardness in water.

(i) 1.00 dm^3 of a hard water sample contains 0.384 g of dissolved magnesium sulfate.

Calculate the number of moles of magnesium sulfate present.

Give your answer in **standard form**. [2]

$$M_r(MgSO_4) = 120$$

$$\text{number of moles} = \frac{\text{mass}}{M_r}$$

Number of moles = ... mol

(ii) 1.00 dm^3 of another hard water sample contains the same mass (0.384 g) of dissolved magnesium chloride.

$$M_r(MgCl_2) = 95$$

State which water sample has more hardness.

Give your reasoning. [2]

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